VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B.Tech. (CSE and its Specialization)	Semester	:	Fall semester 2025-2026
Course Code & Course Title	:	BCSE355L , Cloud Architect Design	Slot	:	B1+TB1
Faculty	:	Dr. PRADEEP K Dr. S VIGNESHWARI Dr. TAMILARASI K Dr. SELLAM V Dr. POORNIMA S Dr. DEEPA NIVETHIKA Dr. RAJESH KUMAR Dr. SUGUNA Dr. LINDA JOSEPH	Class Number	:	CH2025260102018 CH2025260101350 CH2025260102012 CH2025260102007 CH2025260102048 CH2025260101473 CH2025260101673 CH2025260101670 CH2025260101668
Duration	:	90 Minutes	Max. Mark		50

General Instructions: < Use this space to provide additional information such as graph sheet, data book etc.>

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a b	For a multi-tier insurance application that involves a front tier that allows users to interact and see their insurance policy, status, next payment due etc. The Application has a middle tier that processes payments and implements internal business policy. Third tier has the data base for customers, policy, and employee of company. The number of customers of application are increasing, with bonus and coupons are distributed on application. Draw a diagram that shows use of appropriate loadbalancer among multi-tier of this application. (3 Marks) Explain how the chosen loadbalancer helps in this multi-tier application. (3 Marks) Identify any other AWS services that may be used with loadbalancer and role played by the services to make the application high performance from cost and low latency access. (4 Marks)	10	CO1	K3
2		A startup looking to securely store user-uploaded files, some of which must be accessed regularly (like profile images) while others are only needed for occasional reference (such as archived transaction records). The manager of the startup wants a solution to keep costs low by moving older files to a less expensive storage option when they're rarely needed, ensure critical files are	10	CO1	K3

		always quickly retrievable and automatically deletes files after a set retention period for compliance. Describe the decision-making and design steps you would take to architect such a solution in the cloud, detailing how you would organize files, automate transitions between storage types, and enforce the retention policy.			
3		A growing MNC company is designing its new cloud-based learning platform on AWS. The platform will host a high-performance relational database on an EC2 instance that requires low-latency, persistent storage for structured student data. Additionally, the platform will store a large collection of lecture videos, PDFs, and images that need to be accessed concurrently by multiple EC2 instances located in different Availability Zones. Both storage solutions must be durable, secure, and fully managed by AWS.	10	CO3	K3
	a	As the solutions architect, you are required to identify and explain the most suitable AWS storage service for each requirement and justify your answer (7 marks)			
	b	List two key advantages of each choice and mention one limitation for each. (3 Marks)			
4		As the Cloud Architect for a large university, you are tasked with designing a multi-tier student management system on AWS. The system has three layers: a web portal accessed by students and faculty for course registration, grade viewing, and communication; an application layer managing course scheduling, enrolment, and administration; and a database layer securely storing sensitive student records and financial data. Each layer must be network-isolated to optimize performance and enforce strict security. The web portal should be publicly accessible, while the application and database layers must be protected from direct internet access.	10	CO4	K5
	a	The design must ensure high availability and scalability to support peak periods like enrolment deadlines and exam announcements, serving thousands of users without interruption. Deployment across multiple Availability Zones is required for fault tolerance and resilience. Given budget constraints, cost optimization is crucial while maintaining security and performance. Secure access controls, network segmentation, and automated scaling should be incorporated to create a robust, scalable, and cost-effective solution for the university's needs. (3 Marks)			
	b	Explain how you would design the network architecture using public and private subnets and multiple Availability Zones to achieve secure network isolation for the university's student management system. (4 Marks) Explain the role of NAT gateways in the architecture and how they help balance cost-effectiveness and security for the application and database layers.			
	c	Compare AWS Application Load Balancer and Network Load Balancer for this system, focusing on scalability, security, and reliability factors that influence their selection.(3 Marks)			

5

5		John is the IT Head at a multi-specialty hospital that offers telemedicine services to patients across different regions. The hospital's web application allows patients to schedule appointments, access their electronic health records (EHRs) and join live video consultations with doctors. Recently, John noticed that patients in remote areas experience slow load times when accessing medical reports containing large imaging files (e.g., MRI scans) and that live video consultations sometimes lag during peak hours. The hospital must also ensure compliance with healthcare regulations like Health Insurance Portability and Accountability Act (HIPAA), guaranteeing that all patient data remains secure during transmission. In addition, the hospital is planning to expand its telemedicine services internationally, which will require delivering content quickly and securely across multiple continents while handling thousands of concurrent connections without service disruption.	10		
	a	How can John use AWS's global content delivery network to reduce latency for patients in remote areas when accessing large medical imaging files and other health records? (5 Marks)		CO2	K3
	b	Describe how the AWS edge network can help the hospital ensure high availability and performance during peak telemedicine sessions with thousands of simultaneous users. (5 Marks)			

**********All the best **********